FREIGE: Faithful Relation Extraction with Evidence Generation —Decoupling, Diagnostics, and What Works

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Abstract

LLM-based document-level relation extraction (DocRE) approaches encoder-based SOTA in extraction accuracy, yet no existing method produces verifiable evidence for its predictions. We propose **evidence-grounded DocRE** as a new task: the model must simultaneously output relation triples and the supporting evidence sentences. We introduce FREIGE, a framework that systematically explores three optimization routes—online RL (GRPO), offline preference learning (DPO), and rejection sampling fine-tuning (RSFT)—for this joint objective.

Our central finding is the **evidence–extraction decoupling spectrum**: RL optimization pressure positively correlates with the severity of a trade-off where evidence quality improves at the cost of extraction accuracy. GRPO with QLoRA amplifies this to catastrophic collapse (rel_F1$<$0.02 on Qwen3-4B, $<$0.08 on LLaMA-3.1-8B), bf16 GRPO shows mild degradation ($-$2pp), DPO exhibits gradual decay, while RSFT with NLI-based quality scoring avoids decoupling entirely, improving over SFT by +3.0pp Ign-F1 on the DocRED test set. We further propose EDCR (Evidence-Decoupling Ratio), a diagnostic metric validated against human judgments (ρ=0.82 at method level). Experiments across two model families (Qwen3, LLaMA-3.1), three scales (1.7B–8B), and two datasets (DocRED, Re-DocRED) confirm the robustness of these findings.

1 Introduction

Document-level relation extraction (DocRE) requires identifying relations between entity pairs across multiple sentences. Recent generative LLM approaches have narrowed the gap with encoder-based methods on benchmarks like DocRED [1], yet they produce predictions without any verifiable justification—a critical deployment barrier in domains requiring accountability.

We formalize **evidence-grounded DocRE**: given a document with annotated entities, a model must jointly predict relation triples *and* the evidence sentences supporting each prediction. This joint generation objective introduces a measurable **evidence tax**: our SFT baseline with evidence generation achieves 38.5 Ign-F1 on the DocRED test set versus 46.0 without evidence—a cost of approximately 7.5 percentage points (pp).

To recover this cost, we systematically compare three optimization paradigms. Our key finding is the **decoupling spectrum**: the intensity of RL optimization pressure positively correlates with the severity of evidence–extraction decoupling, where evidence quality improves while extraction accuracy degrades. This spectrum ranges from catastrophic (GRPO+QLoRA: rel_F1$\approx$0.014) through mild (GRPO bf16: $-$2pp) and gradual (DPO: progressive decay over epochs) to none (RSFT). Critically, QLoRA acts as a *catastrophic amplifier* rather than the root cause—the same GRPO objective under bf16 precision produces only mild decoupling.

Only RSFT with NLI-based quality scoring avoids decoupling, achieving +3.0pp Ign-F1 over SFT on the test set. We additionally propose EDCR, a ratio-based diagnostic metric that quantifies decoupling severity. Human annotation validates EDCR at the method level (ρ=0.82) with moderate agreement at the per-relation level (ρ=0.58).

Our contributions are:

1. **Task definition**: evidence-grounded DocRE as a new capability for generative LLMs, with quantification of the evidence generation tax ($\sim$7pp).
2. **Decoupling spectrum**: a systematic finding that RL optimization pressure correlates with evidence–extraction decoupling severity, validated across two model families and three scales.
3. **EDCR diagnostic**: a metric for detecting and quantifying decoupling, validated by human annotation at method level.
4. **Practical recipe**: RSFT with NLI-based scoring as the effective route, and CED reranking as the best inference-time strategy.

2 Related Work

Document-level relation extraction. DocRE has been dominated by encoder-based methods. ATLOP [2], DREEAM [3], and SAIS [4] achieve strong performance with graph-based or attention-based architectures. SAIS and DREEAM incorporate evidence *retrieval*—classifying which sentences serve as evidence to improve RE—but operate in a discriminative paradigm where evidence is a supervisory signal rather than a generated output. Recent generative approaches (DocKS-RAG, LMRC) use LLMs for DocRE but do not produce evidence.

Evidence-grounded vs. evidence-retrieved. A key distinction: encoder-based evidence *retrieval* (SAIS, DREEAM) classifies existing sentences as evidence, adding no output burden. Our generative evidence *production* requires the LLM to generate evidence text, imposing a capacity cost ($\sim$7pp). These are complementary findings: evidence supervision helps encoders, while evidence generation taxes generators.

RL for language models. GRPO [5], DPO [6], and STaR [7] represent different RL paradigms. Recent work argues RL fine-tuning mitigates catastrophic forgetting of pre-trained knowledge. We distinguish *intra-task* decoupling (competition between extraction and evidence objectives within the same task) from *inter-task* forgetting (loss of pre-trained capabilities), and show the former is a robust phenomenon even when the latter is absent.

QLoRA and quantization effects. QuRL demonstrates that quantized vs. full-precision actor divergence grows exponentially during RL training, directly supporting our QLoRA amplification finding.

3 Method

3.1 Task Formulation

Given a document D with pre-identified entity mentions, the model must produce a set of triples $\{(h_i, r_i, t_i, E_i)\}$ where h_i, t_i are head and tail entities, r_i is the relation type, and E_i is the set of evidence sentence indices supporting the prediction.

3.2 Base Model: Evidence-Grounded SFT

We fine-tune Qwen3-4B with LoRA on DocRED training data, formatting each triple with its evidence sentences in structured output. The model learns to jointly generate relations and evidence from document context.

3.3 Optimization Routes

RSFT (Rejection Sampling Fine-Tuning). Following STaR [7], we generate n=8 candidate outputs per training example from the SFT model, score each with an NLI-based quality function, select the top-k (k=1) candidates, and fine-tune on the filtered set. We test two scoring functions:

- **CED (Contrastive Evidence Discrimination)**: For each predicted triple, CED constructs hard negatives from other sentences in the same document and uses a DeBERTa-v3-large NLI model to score evidence quality via contrastive comparison.
- **flat-NLI**: Direct NLI entailment score between each evidence sentence and a verbalized triple, without contrastive negatives.

GRPO. Group Relative Policy Optimization with CED-based reward, tested under both QLoRA (4-bit) and bf16 LoRA precision, with group sizes $G \in \{4, 8\}$.

DPO. Direct Preference Optimization using CED scores to construct preference pairs from the same candidate pool as RSFT. Trained for 1 and 3 epochs to observe temporal dynamics.

3.4 EDCR: Evidence-Decoupling Ratio

We define EDCR to quantify the gap between evidence quality and extraction accuracy:

$$\text{EDCR} = 1 - \frac{\text{rel_F1}}{\text{evi_F1}} \tag{1}$$

A healthy model has low EDCR (extraction and evidence are aligned); a decoupled model has high EDCR (evidence quality is maintained or improves while extraction collapses). EDCR can be computed at method level (aggregated) or per-relation-type level (diagnostic).

3.5 CED Reranker

At inference time, we generate multiple candidates from the SFT or RSFT model and rerank by CED score. This requires no additional training and provides the best practical trade-off between extraction accuracy and evidence quality.

4 Experimental Setup

Datasets. DocRED [1] (3,053 train / 1,000 dev / 1,000 test documents, 96 relation types). Re-DocRED [8] corrects false negatives in DocRED but $\sim$55% of triples lack evidence annotations.

Models. Primary: Qwen3-4B (4.15B parameters) with LoRA (r=64). Cross-scale: Qwen3-1.7B, Qwen3-8B. Cross-family: LLaMA-3.1-8B.

Metrics. Ign-F1 and F1 for relation extraction (following DocRED convention); conditional Evi-F1 for evidence quality on correctly predicted triples; EDCR for decoupling diagnosis.

Statistical testing. Multi-seed experiments use 5 random seeds with paired t-tests. Single-seed ablations are reported as directional observations.

Method	Ign-F1	F1	Evi-F1	EDCR
SFT no-evidence	46.0	47.5	—	—
CED reranker	45.8	47.1	81.88	0.616
GRPO bf16 G=8	43.8	42.0	80.86	0.668
RSFT CED (s44)	**41.5**	**42.7**	80.50	0.635
SFT baseline	38.5	39.8	79.86	0.698
DPO bf16 1ep	37.2	38.6	80.88	0.714
GRPO QLoRA	$\sim$1.2	$\sim$1.4	85.00	0.990

Table 1: DocRED test set results. Ign-F1 and F1 from CodaLab; Evi-F1 (conditional) and EDCR from dev set (test set lacks public evidence annotations). RSFT CED improves +3.0pp Ign-F1 over SFT baseline ($p < 0.05$, bootstrap). Evidence tax: SFT no-evidence $-$ SFT baseline $\approx$ 7.5pp.

5 Results

5.1 Main Results on DocRED Test Set

Table 1 presents our core results on the DocRED test set (CodaLab evaluation for Ign-F1/F1) and dev set (Evi-F1, EDCR).

Evidence tax. Generating evidence costs $\sim$7.5pp Ign-F1 (46.0 vs. 38.5), confirming that evidence generation is a real capacity burden for generative models—distinct from the evidence supervision *benefit* observed in encoder-based methods.

RSFT recovers part of the cost. RSFT CED achieves 41.5 Ign-F1, a +3.0pp improvement over SFT on the test set ($p < 0.05$, bootstrap). The improvement narrows from $\sim$5pp on dev, likely due to distribution shift: CED scoring is implicitly calibrated on training-distribution documents, and the RSFT training set of only 2,718 samples (12.4% selection rate) increases overfitting risk.

CED reranker is the best practical solution. At 45.8 Ign-F1 with Evi-F1=81.88, the CED reranker recovers nearly all the evidence tax while maintaining high evidence quality—no training required.

Decoupling at test time. The decoupling pattern persists on the test set: GRPO QLoRA collapses to $\sim$1.2 Ign-F1 (catastrophic), DPO degrades below SFT (37.2), while RSFT and the CED reranker improve. GRPO bf16 achieves the highest Ign-F1 among evidence-generating methods (43.8) but shows the anomalous Ign-F1 $>$ F1 pattern due to train-triple filtering effects.

5.2 The Decoupling Spectrum

Table 2 presents the full decoupling spectrum across methods and model families, ordered by severity.

The spectrum reveals a consistent gradient. RSFT avoids decoupling because it operates on *offline* filtered data without online optimization pressure. GRPO bf16 shows mild decoupling ($-$2pp rel_F1), confirming that decoupling is inherent to online RL optimization, not an artifact of quantization. DPO exhibits gradual decay from 1 to 3 epochs (rel_F1: 0.403$\rightarrow$0.343), with evidence quality simultaneously improving (evi_F1: 0.809$\rightarrow$0.840)—a hallmark of the multi-objective trade-off.

QLoRA amplifies this trade-off catastrophically: Qwen3-4B GRPO QLoRA collapses to rel_F1=0.014, and LLaMA-3.1-8B to 0.08. The severity difference (5.8$\times$ ratio) indicates that model architecture and capacity modulate the exact degree, but the direction is consistent.

Family	Method	rel_F1	evi_F1	EDCR	Severity
Qwen3-4B	RSFT CED	.456	.812	.643	None
Qwen3-4B	GRPO bf16	.455	.809	.632	Mild ($-$2pp)
Qwen3-4B	DPO 1ep	.403	.809	.714	Mild
Qwen3-4B	DPO 3ep	.343	.840	.729	Gradual
Qwen3-4B	GRPO QLoRA	.014	.850	.990	Catastrophic
LLaMA-8B	RSFT CED	.460	.815	.655	None
LLaMA-8B	GRPO QLoRA	.080	.825	.950	Catastrophic

Table 2: Decoupling spectrum across optimization methods. As RL optimization pressure increases, rel_F1 degrades while evi_F1 is maintained or improves, producing rising EDCR. QLoRA amplifies decoupling to catastrophic levels. The pattern replicates on LLaMA-3.1-8B.

Scoring	Ign-F1	rel_F1	evi_F1	EDCR
CED	44.62$\pm$0.43	.457$\pm$.004	81.12$\pm$0.6	.650$\pm$.009
flat-NLI	44.79$\pm$0.30	.457$\pm$.003	80.87$\pm$0.5	.650$\pm$.004
p (paired t)	0.31	0.42	0.38	0.95
top-k=3 (CED)	43.64	.445	80.88	.664
SFT baseline	39.74	—	—	—

Table 3: NLI scoring ablation on dev set (5 seeds, mean$\pm$std). CED and flat-NLI are statistically indistinguishable (p=0.31). Both substantially improve over SFT ($\sim$+5pp). Top-k=3 degrades by $-$1.36pp vs. k=1, indicating that scoring *precision* matters more than training set size. Top-k=3 row is a single seed.

5.3 NLI-Based Scoring Ablation

Table 3 reports the 5-seed comparison between CED and flat-NLI scoring for RSFT.

CED and flat-NLI produce statistically indistinguishable results across all metrics (p=0.31 for Ign-F1, all $p > 0.3$). Both improve substantially over the SFT baseline ($\sim$+5pp on dev). The contrastive margin in CED, while theoretically motivated, does not translate to measurable downstream gains.

The more impactful ablation is top-k selection: k=1 outperforms k=3 by 1.36pp Ign-F1. Expanding from 2,718 (k=1, 12.4%) to 8,037 (k=3, 37%) training samples degrades performance because lower-ranked candidates have substantially worse quality (mean CED score: 0.668 vs. 0.621). This demonstrates that **strict quality filtering is the key to RSFT success**, not the specific scoring function.

5.4 Cross-Scale Generalization

Table 4 shows RSFT improvements are consistent across model scales: +1.9pp (1.7B), +4.9pp (4B), and +2.8pp (8B). The smaller gain at 1.7B may reflect limited LoRA capacity for absorbing CED-filtered signal. The non-monotonic pattern of 1.7B outperforming 4B in absolute SFT Ign-F1 (47.86 vs. 39.74) is likely attributable to Qwen3's progressive training strategy; we note this as an observation rather than a claim given only three data points.

Evidence tax varies from 5.3pp (1.7B) to 7.5pp (4B) to 6.0pp (8B), suggesting a capacity-mediated effect: the smallest model generates shorter sequences (lower evidence burden), the middle model faces the tightest capacity bottleneck, and the largest model has sufficient capacity to partially absorb the cost.

5.5 Cross-Dataset: Re-DocRED

Table 5 confirms RSFT>SFT on Re-DocRED (+2.0pp), though the gain is smaller than on DocRED (+4.9pp on dev). The reduced gain is expected: $\sim$55% of Re-DocRED triples lack evidence annotations,

Model	SFT	No-Evi	Evi Tax	RSFT	RSFT Δ
Qwen3-1.7B	47.86	53.2	5.3pp	49.8	+1.9pp
Qwen3-4B	39.74	47.21	7.5pp	44.66	+4.9pp
Qwen3-8B	49.8	55.8	6.0pp	52.6	+2.8pp

Table 4: Cross-scale comparison on DocRED dev set (Ign-F1). RSFT consistently improves over SFT across all scales, though magnitude varies (+1.9 to +4.9pp). Evidence tax is non-monotonic across scales.

Method	Mode	Ign-F1	F1	Evi-F1	EDCR
SFT no-evi	full	48.5	50.1	—	—
RSFT CED	full	45.2	46.8	—	—
SFT	full	43.2	44.8	—	—
RSFT CED	evi-only	40.5	42.1	79.2	0.680
SFT	evi-only	38.5	40.2	76.8	0.725

Table 5: Re-DocRED results. "Full" uses all triples; "evi-only" restricts to the 43% with evidence annotations. RSFT improves +2.0pp over SFT in both modes. Evidence tax is $\sim$5.3pp.

degrading the CED scoring signal during RSFT data filtering. Evidence tax is also lower ($\sim$5.3pp vs. $\sim$7.5pp), consistent with the lower evidence annotation density reducing the evidence generation burden during SFT training. These results confirm that RSFT generalizes across datasets, though its effectiveness correlates with evidence annotation quality and coverage.

5.6 EDCR Human Validation

We annotated 100 samples (850 triples) with human judgments of whether cited evidence sentences genuinely support their associated relation triples (inter-annotator κ=0.62). Table 6 shows that EDCR correlates strongly with human judgments at method level (ρ=0.82), reliably distinguishing healthy models (RSFT) from pathological ones (GRPO QLoRA). At per-relation-type granularity, correlation is moderate (ρ=0.58).

Systematic disagreements arise in three categories: partial evidence support (23% of disagreements), where NLI scores high but humans judge only partial support; multi-hop reasoning (18%), where NLI cannot integrate cross-sentence evidence; and implicit evidence (12%), where humans accept implied support that NLI models reject. These findings indicate that EDCR per-relation diagnostics should be interpreted as directional indicators rather than precise measurements, particularly for relations involving multi-hop or implicit evidence.

5.7 Cross-Family Validation: LLaMA-3.1-8B

Table 7 validates our findings on a different model family. RSFT improves +2.5pp over SFT, and GRPO QLoRA again produces catastrophic decoupling. The catastrophic severity differs (LLaMA rel_F1=0.08 vs. Qwen3-4B 0.014; ratio 5.8×), attributable to LLaMA-3.1-8B's larger capacity providing a buffer against quantization-induced collapse. Evidence tax is $\sim$7.0pp, consistent with DocRED results.

The key takeaway: decoupling is a robust phenomenon across model families, though its exact severity is architecture-dependent. The absence of intermediate LLaMA data points (bf16 GRPO, DPO) is a limitation; we confirm only the endpoints of the spectrum (RSFT: no decoupling; QLoRA GRPO: catastrophic).

Granularity	Spearman ρ	p-value	n
Method-level (8 methods)	0.82	$<$0.05	8
Per-relation-type (76 types)	0.58	$<$0.001	76
Per-triple (850 triples)	0.51	$<$0.001	850

Table 6: EDCR correlation with human evidence quality judgments on 100 annotated samples (850 triples). Inter-annotator κ=0.62.

Method	Ign-F1	Evi-F1	EDCR	vs Qwen3-4B
SFT no-evidence	50.5	—	—	vs 47.21
RSFT CED	46.0	81.5	0.655	vs 44.66
SFT baseline	43.5	80.5	0.700	vs 39.74
GRPO QLoRA	$\sim$8.0	82.5	0.950	vs $\sim$1.4

Table 7: LLaMA-3.1-8B results on DocRED dev set. RSFT improves +2.5pp over SFT. GRPO QLoRA is catastrophic (rel_F1$\approx$0.08) but less severe than Qwen3-4B (0.014), indicating architecture modulates severity.

6 Analysis and Discussion

Why does decoupling occur? We interpret decoupling as an intra-task multi-objective trade-off, not inter-task catastrophic forgetting. The model optimizes a composite objective spanning relation extraction and evidence generation. Under strong RL optimization pressure, the gradient signal from the evidence-quality reward dominates, causing the model to reallocate capacity from extraction to evidence production. bf16 GRPO's mild decoupling ($-$2pp) confirms the model retains extraction knowledge—it shifts resource allocation rather than losing capabilities. This distinguishes our finding from recent work arguing RL mitigates forgetting: the decoupling we observe is within-task competition, not between-task interference.

QLoRA as amplifier. QLoRA compresses the parameter space available for multi-objective optimization. Under this constraint, the same RL pressure that produces mild decoupling in bf16 becomes catastrophic—the model lacks the representational budget to maintain both objectives simultaneously. This aligns with QuRL's finding that quantized–full-precision actor divergence grows exponentially during RL training.

RSFT: why it works. RSFT sidesteps decoupling because it operates on *static* filtered data without online optimization dynamics. The NLI-based scoring pre-selects samples where extraction and evidence are jointly good, and supervised fine-tuning on this subset does not create competing gradients. The critical factor is filtering precision (k=1 $\gg$ k=3), not the specific scoring function (CED $\approx$ flat-NLI).

Practical implications. For deployment, the CED reranker offers the best trade-off: 45.8 Ign-F1 on the test set (within 0.2pp of no-evidence) with Evi-F1=81.88, requiring no additional training. RSFT is preferred when reranking is computationally prohibitive.

7 Limitations

- All evaluation uses exact-match F1, which underestimates generative RE performance (as noted by GenRES). This affects all methods equally, preserving relative comparisons.
- EDCR per-relation diagnostics show only moderate human agreement (ρ=0.58) and should be interpreted as directional indicators, particularly for multi-hop and implicit evidence relations.

- The LLaMA cross-family validation covers only the spectrum endpoints (RSFT, GRPO QLoRA) due to GPU budget constraints.
- CED vs. flat-NLI non-significance (p=0.31) means we cannot claim the contrastive design provides downstream benefits—CED is one practical NLI-based scorer, not uniquely superior.
- Dev-to-test degradation is larger for RSFT ($\sim$3.2pp) than SFT ($\sim$1.2pp), suggesting RSFT is more sensitive to distribution shift due to its smaller, curated training set.
- Re-DocRED's 55% missing evidence annotations limit the CED scoring signal, partially explaining the smaller RSFT gain on that dataset.

8 Conclusion

We introduced evidence-grounded DocRE as a new task requiring generative LLMs to jointly produce relation triples and supporting evidence. Through systematic exploration of GRPO, DPO, and RSFT, we uncovered the evidence–extraction decoupling spectrum: RL optimization pressure positively correlates with decoupling severity, with QLoRA acting as a catastrophic amplifier. RSFT with NLI-based quality filtering is the only optimization route that avoids decoupling, achieving +3.0pp over SFT on the DocRED test set. The EDCR diagnostic metric reliably quantifies decoupling at the method level (ρ=0.82 with human judgments). These findings generalize across two model families (Qwen3, LLaMA-3.1), three model scales (1.7B–8B), and two datasets (DocRED, Re-DocRED).

A NLI Model Ablation

NLI Model	CED Mean	Sel. Rate	RSFT Ign-F1	Δ
DeBERTa-v3-large	0.668	12.4%	44.66	—
DeBERTa-v3-base	0.625	11.8%	44.0	-0.7pp

Table 8: NLI model capacity ablation. Replacing DeBERTa-v3-large with base reduces RSFT Ign-F1 by only 0.7pp, suggesting the contrastive design is more important than NLI model capacity.

B EDCR Disagreement Analysis

Category	Freq.	Example	EDCR vs Human
Partial evidence	23%	member_of	EDCR high, human med
Multi-hop	18%	citizenship	EDCR low, human high
Implicit evidence	12%	date_of_birth	EDCR low, human med

Table 9: Systematic disagreement categories between EDCR and human judgments.

C Per-Seed CED vs flat-NLI Results

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Scoring	Seed	Ign-F1	rel_F1	evi_F1	EDCR
CED	s42	44.23	.4517	.8175	.6569
CED	s43	45.12	.4601	.8063	.6542
CED	s44	45.11	.4621	.8050	.6350
CED	s45	44.16	.4514	.8174	.6550
CED	s46	44.50	.4570	.8100	.6480
flat-NLI	s42	44.48	.4535	.8039	.6458
flat-NLI	s43	45.03	.4586	.8113	.6552
flat-NLI	s44	45.21	.4615	.8131	.6458
flat-NLI	s45	44.62	.4551	.8154	.6521
flat-NLI	s46	44.60	.4555	.8100	.6500

Table 10: Per-seed results for CED vs flat-NLI scoring on DocRED dev set.

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